

Going forward, will the equity premium continue to be high? Or will the next ten years be another Lost Decade for stocks? To answer these questions, we must understand what factors explain equity return performance and volatility and ascertain whether these risk factors will persist in the future. These are issues of how the *aggregate market* is priced and moves over time.

2. The Equity Risk Premium

In previous chapters we've described how risk premiums arising from factor risks compensate investors for bearing losses during bad times. One of those factors, described in chapter 7, was *consumption*—which was part of a broader set of variables capturing economic growth in general. In a *consumption-based asset pricing model*, all risk is summarized by consumption—bad times occur when society, or all agents in the economy, are consuming less. We summarize all agents in society by assuming a *representative agent*, which we can think about as the average person or investor. To compensate investors for holding equities, the average investor earns an *equity risk premium*. In the most basic consumption models, there is only one factor, per capita real consumption growth, and other factors matter only to the extent they affect consumption. This makes sense because ultimately we care about what is finally consumed by agents in the economy.

The *equity premium puzzle* is that, using consumption as a risk factor, the equity premium should be very modest. How modest? In 1985, a paper written by Rajnish Mehra and Edward Prescott challenged the profession. They claimed that the equity premium over risk-free assets should be well below 1% for reasonable levels of risk aversion, which are levels of risk aversion between 1 and 10. These are the risk aversion levels that most individuals have (see chapter 2). In contrast, historically the equity premium has been very high. Over 1900–2010, the Dimson, Marsh, and Staunton (2011) data reported in Figure 8.2 show that the average return of U.S. equities in excess of bonds was 6.2% and in excess of bills was 7.4%. The world equity premium in excess of bonds was 5.0% and in excess of bills was 6.0%. Thus, equities have historically had much higher returns than predicted by simple economic models.

Mehra and Prescott titled their paper, “The Equity Premium: A Puzzle,” and it quickly spawned a new literature.¹ They based their model on Lucas (1978), which is one of the most influential papers in financial economics and was a factor in Robert Lucas's Nobel Prize in 1995. Prescott himself would go on to win the Nobel in 2004 for advancements in macroeconomic modeling.

¹A recent summary is Mehra (2006).